

EXPECTED TECHNICAL & PROJECT QUESTIONS & STAR ANSWERS

Q1 (Technical): Walk me through how you handle data cleaning and feature engineering in a machine learning pipeline.

- **Situation:** Raw dataset in the Adaptive Assessment project contained missing values and unscaled feature metrics.
- **Task:** Prepare clean, normalized input vectors for Random Forest classification.
- **Action:** Implemented Pandas pipelines for missing value imputation, encoded categorical variables, and extracted relevant feature subsets using Scikit-learn.
- **Result:** Improved classification accuracy and produced clean, reproducible feature sets.

Q2 (Project-Based): How do you document and share your Machine Learning results?

STAR Answer:

- **Situation:** Developing the NeuraSight AI research project and UNESCO EchoBreaker simulator.
- **Task:** Document findings for both technical peers and general users.
- **Action:** Created Jupyter Notebooks for model evaluation, built interactive web dashboards (HTML/JS, Firebase), compiled formal LaTeX reports, and published everything on GitHub.
- **Result:** Delivered complete, multi-layer research projects accessible to researchers and developers.